Pierre Onghena

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EXPERIENCE

Mines Paris – PSL Oct 2023 – Dec 2026
PhD Researcher *Paris, France*

- Developed deep learning methods for point clouds in PyTorch, trained on the Jean Zay HPC cluster with Slurm.
- Established a new state of the art in unsupervised 3D keypoint detection (GEKey, BMVC 2026 Oral).
- Co-developed a femoral landmark detector with lower error than expert annotators (LmPT, ISBI 2026 Oral).
- Developed a morphological skeletonization method for informed point sampling (MorphoSkel3D, 3DV 2025).
- Mentored master's students in research internships and served as teaching assistant for a deep learning course.

ESCP Business School 2024 – 2026
Lecturer *Paris, France*

- Led lectures for Bachelor in Management students, encouraging participation and conceptual understanding.
- Taught mathematics in the first semester to equip students with analytical and quantitative reasoning skills.
- Taught statistics in the second semester to prepare students to reason about uncertainty in complex problems.

IBM May – Sep 2023
Data Engineer *Brussels, Belgium*

- Adapted large language models to domain-specific document collections for corporate and public-sector clients.
- Contributed to consulting engagements and technical tenders within IBM's global technology practice.
- Translated machine learning expertise into client-facing value propositions and solution proposals.

The Cross Product Jul 2022 – Mar 2023
R&D Engineer *Fontainebleau, France*

- Built LiDAR perception for rail, road, and industrial infrastructure, working with technical and business teams.
- Delivered results that led to a peer-reviewed publication and conference presentation (ICCVW 2023).

PUBLICATIONS (*equal contribution)

"GEKey: Learning Geodesic Eccentric 3D Keypoints via Self-Supervision" **BMVC 2026 Oral**
Pierre Onghena, Santiago Velasco-Forero, Beatriz Marcotegui *Lancaster, UK*

"LmPT: Conditional Point Transformer for Anatomical Landmark Detection on 3D Point Clouds" **ISBI 2026 Oral**
Matteo Bastico*, **Pierre Onghena***, David Ryckelynck, Beatriz Marcotegui, *London, UK*
Santiago Velasco-Forero, Laurent Corté, Caroline Robine-Decourcelle, Etienne Decencière

"MorphoSkel3D: Morphological Skeletonization of 3D Point Clouds for Informed Sampling" **3DV 2025**
Pierre Onghena, Santiago Velasco-Forero, Beatriz Marcotegui *Singapore*

"Rotation-invariant Hierarchical Segmentation on Poincaré Ball for 3D Point Cloud" **ICCVW 2023**
Pierre Onghena, Leonardo Gigli, Santiago Velasco-Forero *Paris, France*

"Comparing User Perception of Explanations Developed with XAI Methods" **WCCI 2022**
Jonathan Aechtner, Lena Cabrera, Dennis Katwal, **Pierre Onghena**, Diego Penroz Valenzuela, Anna Wilbik *Padua, Italy*

EDUCATION

PhD in Computer Vision, Mines Paris – PSL, Centre for Statistics and Images 2023 – 2026
Thesis: "Geometric Priors for Deep Learning on 3D Point Clouds"
Advisors: Beatriz Marcotegui & Santiago Velasco-Forero

MSc Artificial Intelligence, Maastricht University 2020 – 2022
Exchange semester: Paris Dauphine – PSL (MSc Artificial Intelligence, Systems & Data) Jan – Jun 2022

MSc & BSc Business and Information Systems Engineering, KU Leuven 2019 – 2020

BSc Business Engineering, KU Leuven 2016 – 2019